

Year 9 Science – Booklet 4: Physics

Work & Energy (Energy Transfers and Sound) — ANSWERS

Quick Test – p.93 (Energy changes)

1. fuel + oxygen → carbon dioxide + water + energy
2. Combustion is fast – once a fuel is ignited it reacts almost instantly, releasing its energy quickly (often producing a flame). Respiration is much slower – it releases energy from food gradually inside living cells, without a flame, at a controlled rate.
3. Radiation is the transfer of heat energy as (infrared) waves. It does not need particles, so it can travel through a vacuum, and the objects involved do not need to be touching.
4. 1000 J – by conservation of energy, all of the gravitational potential energy could be transferred to kinetic energy (assuming no energy is wasted, e.g. to friction or air resistance).

Quick Test – p.95 (Sound)

1. It means 50 complete sound waves are produced (pass a given point) every second.
2. Sound waves travel out from a source, hit a surface, and are reflected back. If this reflected sound reaches your ear a short time after the original sound, you hear it separately as an echo.
3. Cover the walls and ceiling with soft, angled, sound-absorbing tiles/foam, so that sound waves keep reflecting within the room until their energy has been absorbed, instead of reflecting straight back into the room.
4. An electrical signal is sent to the loudspeaker, causing electromagnets to move a fabric skin/membrane (diaphragm) back and forth. This makes the surrounding air particles vibrate, creating a sound wave that travels to the ear.

Vocabulary Builder (39 marks)

1. Complete the sentences

- a). A cup of hot tea has a greater store of **thermal** energy than a cup of cold tea.
- b). A fast-moving car has a greater store of **kinetic** energy than a slow-moving car of the same mass.
- c). A 2 kg bag of sugar has a greater store of **chemical** energy than a 1 kg bag of sugar.
- d). A boy's hands are cold so he rubs them together transferring **chemical** energy stored in his muscles to the **thermal** energy of his hands.
- e). When a moving car's brakes are applied, **kinetic** energy of the car's wheels is transferred to **thermal** energy in the brakes.
- f). As a car pulls away from traffic lights, its store of **chemical** energy decreases, and its store of **kinetic** energy increases.

2. Does the activity involve doing work?

- a). Pushing a shopping trolley along the aisle at the supermarket – **Yes**
- b). Trying to lift an object that is too heavy to be moved – **No** (no movement occurs, so no work is done, even though a force is applied)
- c). Warming your hands on a hot cup of coffee – **No** (this is a thermal energy transfer, not a force causing movement)

3. True or false (man pushing pushchair at steady speed)

- The man is doing work. – **True**
- The man's store of chemical energy in his muscles is decreasing. – **True**
- The pushchair's store of kinetic energy is increasing. – **False** (steady speed means constant kinetic energy)

4. **Catapult released – what happens to the stores of energy?**

The store of elastic potential energy in the stretched rubber band decreases as it is transferred into kinetic energy of the rocket, which flies off (a small amount is also wasted as heat and sound).

5. **Tennis player throwing a ball vertically**

In throwing the ball, the tennis player does **work** and transfers **kinetic** energy to the ball. As the ball moves upwards, its store of **kinetic** energy decreases and its store of **gravitational potential** energy increases until it reaches its maximum height.

6. **Hot tea poured into a cold cup**

- The thermal energy store of the tea decreases while the thermal energy store of the cup increases, until both reach the same temperature (thermal equilibrium).
- The temperature difference between the tea and the cup drives the transfer (heat flows from the hotter object to the cooler object by conduction, until they are equal).
- Conduction.
- It will increase slightly, as some thermal energy from the hot tea/cup is transferred to the surrounding air.

7. **Chemical energy stored in car fuel**

- Petrol.
- ...pollute the air.
- ...such as **coal** (or natural gas), producing gases that cause climate **change**.

8. **Tuning fork**

- Vibration.
- Air.
- It makes the eardrum vibrate, at the same frequency as the sound wave.
- A longitudinal wave.

9. **Loudspeaker**

- The electric current makes the **diaphragm** of the loudspeaker vibrate. This vibration makes the air **particles** vibrate, creating a sound wave. If the vibrations are speeded up, the sound has a higher **frequency**.
- Hertz.
- Hz.

Maths Skills (17 marks)

1. **work done = force × distance**

- i). $5 \text{ N} \times 20 \text{ m} = \mathbf{100 \text{ J}}$
- ii). $10 \text{ N} \times 25 \text{ m} = \mathbf{250 \text{ J}}$
- iii). $50 \text{ N} \times 0.8 \text{ m} = \mathbf{40 \text{ J}}$
- iv). $600 \text{ N} \times 2 \text{ m} = \mathbf{1200 \text{ J}}$
- The store of **chemical** energy in the person's muscles decreases and the person's store of **gravitational potential** energy increases.

- c) i). $8000 \text{ N} \times 80 \text{ m} = \mathbf{640\,000 \text{ J}}$
 c) ii). $640\,000 \div 1000 = \mathbf{640 \text{ kJ}}$
 d). $100\,000 \text{ N} \times 3000 \text{ m} = \mathbf{300\,000\,000 \text{ J} = 300\,000 \text{ kJ}}$

2. Gardener, plank and boulder

- a). $400 \text{ N} \times 0.6 \text{ m} = \mathbf{240 \text{ J}}$
 b). Work in = work out (no friction), so height = $240 \text{ J} \div 1200 \text{ N} = 0.2 \text{ m}$. Tick **0.2 m**.

3. Chemical energy in a litre of petrol

$800 \text{ g} \times 45\,000 \text{ J/g} = \mathbf{36\,000\,000 \text{ J}}$ ($3.6 \times 10^7 \text{ J}$)

4. Chemical energy per gram of chocolate

- $2200 \text{ kJ} \div 100 = \mathbf{22 \text{ kJ}}$
- $22 \text{ kJ} = \mathbf{22\,000 \text{ J}}$

5. Speed of sound (Maisie's echo estimate)

Sound travels to the wall and back: distance = $150 \text{ m} + 150 \text{ m} = 300 \text{ m}$, in a time of 1 s. speed = distance $\div$ time = $300 \text{ m} \div 1 \text{ s} = \mathbf{300 \text{ m/s}}$.

6. 20 kHz in Hz

$20 \text{ kHz} = \mathbf{20\,000 \text{ Hz}}$

Testing Understanding (19 marks)

1. Shopping trolley

- a). Tick **"Pushing a full trolley along the full length of the aisle."** (greatest force over the greatest distance = most work)
 b). While pushing the trolley, the store of **chemical** energy in your muscles decreases. As you push it at a steady speed, you are doing work against friction, so the store of **thermal** energy in the trolley's wheels increases.

2. Ramp and fridge

- a). work done = force $\times$ vertical height lifted = $1000 \text{ N} \times 0.5 \text{ m} = \mathbf{500 \text{ J}}$
 b). Work is the same either way (ignoring friction): force = $500 \text{ J} \div 4.0 \text{ m} = 125 \text{ N}$. Tick **125 N**.
 c). • Chemical energy in Isla's muscles is transferred to gravitational potential energy – **True**
 • Isla would be doing the same amount of work whether she uses the ramp or not – **True**
 • Isla has to exert a greater force getting the fridge into the van when she uses the ramp – **False** (the ramp reduces the force needed – that is the point of using it)

3. Samara's thrown ball (points X, Y, Z)

Most gravitational potential energy at point **Y** (the highest point). Least gravitational potential energy at **X and Z** (both at ground level). Most kinetic energy at **X and Z** (both at ground level, where all the energy is kinetic).

4. Cyclist freewheeling down a hill (A→B→C→D)

- a). **A to B** (the cyclist is accelerating/speeding up here).
 b). **D** (he has come to a complete stop, so all of the energy that started as gravitational potential energy has ended up as thermal energy by this point).
 c). **B to C** (speed – and so kinetic energy – stays constant here, so all the gravitational potential energy lost in this section must have been converted directly to thermal energy).

5. Electric bell in a container being evacuated

Sound needs a medium (particles) to travel through. As the vacuum pump removes air from the container, there are fewer and fewer particles to carry the sound energy from the bell to the outside, so less sound energy is transferred and the sound gets quieter. The hammer can still be seen hitting the gong because light (unlike sound) does not need a medium to travel. If a total vacuum were reached, no sound at all would be heard.

6. Speed of sound from two microphones (1.55 m apart, 0.00453 s)

speed = distance $\div$ time = $1.55 \text{ m} \div 0.00453 \text{ s} = 342 \text{ m/s}$ (to 3 s.f.)

Working Scientifically (15 marks)

1. Jade's insulation experiment

a). The thickness of the insulating layer affects how quickly heat is lost, so it must be kept the same for all three tests (a controlled variable) – otherwise any difference in cooling rate could be caused by different thicknesses rather than by the insulating material itself, making the comparison unfair. The apparatus achieves this automatically because the gap between the fixed inner and outer beakers is a constant size, so each material fills the same-sized gap.

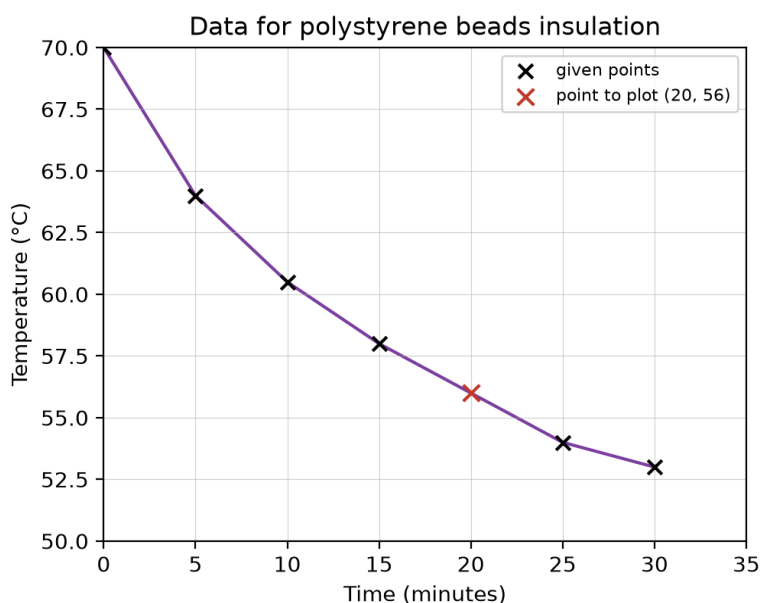
b). She should use the **measuring cylinder**. Measuring cylinders are designed with fine graduations for accurately measuring/reading a specific volume of liquid, whereas the volume markings on a beaker are much less precise – using the cylinder ensures 250 ml is measured accurately and consistently each time, keeping the test fair.

2. Temperature-fall table and insulating order

Temperature fall in 30 minutes: polystyrene beads = $70 - 53 = 17^\circ\text{C}$; cotton wool = $70 - 55 = 15^\circ\text{C}$; sawdust = $70 - 50 = 20^\circ\text{C}$.

Insulating order (best insulator first – smallest temperature fall first): 1. **Cotton wool** 2. **Polystyrene beads** 3. **Sawdust**.

3. Graph – plot the two remaining points and draw a curve



The two points still to plot are (20 min, 56°C) and (25 min, 54°C) – shown in red above. All 7 points should be joined with a single smooth curve (not straight lines), sloping steeply down at first and levelling off as time passes.

4. As time passes, does the temperature drop more quickly or more slowly?

More slowly. The temperature fall each 5-minute interval gets smaller over time (6, 3.5, 2.5, 2, 2, 1°C), so the rate of cooling decreases as time passes.

Science in Use (8 marks)

1. Uses of sound – jewellery cleaner (40 kHz) and dental plaque remover (37 kHz)

- a). Human hearing range: from **20 Hz** to **20 000 Hz**.
- b). Ultrasound is sound with a frequency higher than the upper limit of human hearing (above about 20 000 Hz / 20 kHz) – too high-pitched for a human to hear.
- c). **Yes**, both count as ultrasound – the jewellery cleaner (40 kHz) and the dental plaque remover (37 kHz) both produce sound above 20 kHz, the upper limit of human hearing.
- d). Sound waves reflect off boundaries between different tissues/materials inside the body; detecting these reflections is what allows the ultrasound image to be built up.
- e). Sound travels fastest through solids (steel, then aluminium), slower through the liquid (water), and slowest through the gas (air) – i.e. speed generally decreases from solid → liquid → gas, because particles are progressively further apart and less strongly bonded, so energy transfers between them more slowly.

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